

From Collective Intelligence to Consciousness

WHERE DOES EXPERIENCE BEGIN?

A journey through life, information, organization — and the biggest question of all.

"Perhaps the most difficult question is no longer 'What is life?' but 'What does it mean for something to experience being alive?'"



A JOURNEY SO FAR

In Articles 1–8, we explored how biology may be more than chemistry — moving from molecules to collectives.



1 INTELLIGENCE IS NOT CONSCIOUSNESS

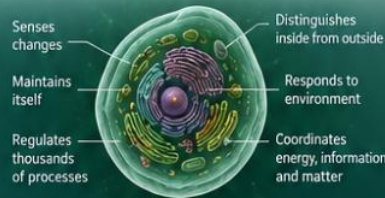
A system can process information, make decisions and adapt — without necessarily experiencing anything.



A thermostat responds A plant adapts An immune cell remembers But do they experience?

2 LIFE IS AN ORGANIZED PROCESS

A living cell is not just a collection of molecules. It is a dynamic, self-maintaining system that senses, responds and regulates.



Life is not passive matter.
A living system is continually doing something about its own existence.

3 WHEN MANY BECOME ONE

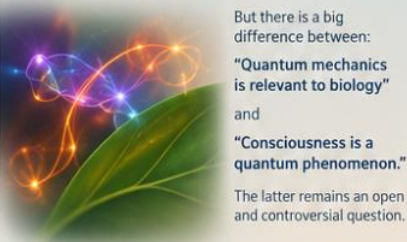
Interactions can create new properties that no individual member has.



Organization creates something new.
But emergence does not automatically mean consciousness.

4 COULD QUANTUM PROCESSES PLAY A ROLE?

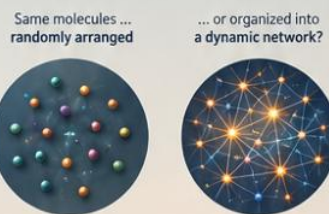
Quantum effects are now well documented in several biological systems (e.g., photosynthesis, vision, magnetoreception).



But there is a big difference between:
"Quantum mechanics is relevant to biology"
and
"Consciousness is a quantum phenomenon."
The latter remains an open and controversial question.

5 IS IT ABOUT ORGANIZATION?

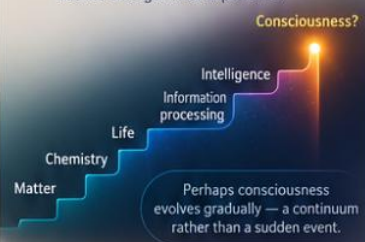
Maybe consciousness depends less on the ingredients and more on how they are organized.



Chemically similar. Functionally different.
Could consciousness be a property of a particular kind of organization?

6 A PHASE TRANSITION?

As systems become more integrated, self-referential and information-rich, could there be a qualitative transition — not just more computation, but the emergence of experience?



7 WHAT WOULD WE LOOK FOR?

If consciousness is a property of organization, we might look for signatures such as:

- ✓ How strongly is information integrated?
- ✓ How much of the system participates in a global state?
- ✓ How quickly can information propagate?
- ✓ How much memory does it retain?
- ✓ How does it distinguish internal from external signals?
- ✓ Does it maintain a model of itself?
- ✓ How does its organization change with stress or development?



BIG QUESTIONS AHEAD

- Could there be degrees of consciousness?
- Where does experience begin — at the neuron, network, brain, organism, or collective?
- Could a non-biological system ever be conscious?
- If we build an artificial system, how would we know if it truly experiences something?

Perhaps consciousness is not a destination, but a landscape — and we are still mapping it.



In the last article of this series, I asked a deceptively simple question:

Can a collective think?

The question arose naturally from the journey we have taken so far.

We began by asking whether biology is more than chemistry. We then looked at the cell not as a bag of molecules, but as an extraordinarily organized system. We asked whether a cell can compute, whether biological systems process information, whether quantum phenomena might have a role in biology, and whether biological computation has anything in common with artificial intelligence.

Then came a more uncomfortable question:

Is intelligence older than the brain?

The answer may be yes—if we are willing to use the word *intelligence* carefully.

A bacterium can sense its surroundings and change its behaviour. Plants can respond to their environment and adjust their growth. Immune cells can recognize, remember and respond. Groups of organisms can collectively solve problems that no individual member could solve alone.

None of this means that a bacterium or a plant is “thinking” in the human sense.

And that brings us to the next question.

If intelligence does not necessarily require a brain, does consciousness require one?

This is where the story becomes much more difficult.

Intelligence is not consciousness

We should pause here, because these two words are often used as though they mean the same thing.

They do not.

A system can process information, make decisions and adapt to its environment without there being any evidence that it *experiences* anything.

A thermostat responds to temperature. A computer processes information. A plant changes its growth in response to light. An immune system distinguishes between molecular patterns and mounts appropriate responses.

We would not normally say that a thermostat *feels* hot or that an immune cell *experiences* an infection.

So there is a gap between **information processing** and **experience**.

And perhaps that gap is the most interesting gap in biology.

We have become reasonably comfortable with the idea that matter can organize itself into systems that process information. We are becoming increasingly comfortable with the idea that such systems can display adaptive behaviour.

But how does a system get from processing information to having an inner point of view?

How does *something happening* become *something being experienced*?

That is the mystery.

Perhaps the brain is not the beginning

Our traditional picture is straightforward.

There is matter.

Matter became chemistry.

Chemistry became life.

Life became increasingly complex.

Eventually nervous systems appeared.

Eventually brains appeared.

And somewhere inside the brain, consciousness emerged.

There is nothing inherently wrong with this picture. In fact, much of modern neuroscience is built upon it.

But the journey through this series has made me wonder whether we are looking at the problem from the wrong end.

Perhaps consciousness is not something that suddenly appears when a sufficiently complicated brain arrives.

Perhaps the important story begins much earlier—with **organization**.

A living cell is already doing something remarkable.

It is continuously distinguishing inside from outside. It is maintaining itself. It is sensing changes. It is responding. It is regulating thousands of molecular processes. It is coordinating energy, information and matter.

None of this demonstrates consciousness.

But it does tell us something important.

Life is not passive matter.

A living system is continually *doing something about its own existence*.

That distinction may matter.

The cell is more than the sum of its molecules

This was one of the central ideas of the earlier articles in this series.

If we take a cell apart, we can identify proteins, lipids, nucleic acids, ions and metabolites. We can describe their structures and chemical reactions.

But something is lost when we reduce the cell to the list of its components.

The cell is also an **organized process**.

Its molecules are connected through networks. Signals move through pathways. Energy flows through compartments. Molecular machines operate in coordinated ways. Feedback loops regulate activity. Spatial organization determines which molecules can interact and when.

The cell is therefore not merely *made of* molecules.

It is also a particular **way in which molecules are organized**.

That distinction has quietly followed us through this entire series.

Perhaps it is becoming even more important now.

Because consciousness may not depend simply on *what matter is present*.

It may depend on **how matter is organized and how information flows through that organization**.

When many become one

Article 8 took us from individual organisms to collectives.

This raises another fascinating possibility.

Consider a colony, a microbial community, an immune system, or even a developing organism. Individual components interact continuously. Information is exchanged. Local decisions influence neighbouring components, and those interactions can produce behaviour at a larger scale.

The collective can therefore acquire properties that are difficult to assign to any individual component.

A single neuron is not a mind.

Yet billions of neurons interacting in the right way give rise to a brain.

A single ant has limited behavioural possibilities. An ant colony can organize extraordinarily complex activities.

A single immune cell does not constitute an immune system.

A single molecule does not constitute a cell.

At every level, organization creates something new.

But there is an important warning here.

Emergence does not automatically mean consciousness.

We cannot simply say that when a system becomes sufficiently complex, consciousness appears.

That would only move the mystery one step further.

We would still have to explain *why* a particular kind of organization should produce experience.

And this is where my earlier quantum questions return

This is also where some of the ideas I have explored in my earlier work come back into the discussion.

I have previously asked whether quantum phenomena might play a deeper role in biological organization.

Quantum biology itself is no longer merely a philosophical curiosity. There is serious experimental work examining quantum effects in biological systems, including processes involving photosynthesis, vision, magnetoreception and molecular reactions. The broader field is now explicitly asking whether quantum phenomena can have functional significance in biological systems.

But there is an enormous difference between saying:

“Quantum mechanics is relevant to biology.”

and saying:

“Consciousness is a quantum phenomenon.”

The first statement is firmly within modern physics.

The second remains an open and controversial hypothesis.

This distinction is important.

Some of my earlier writings explored ideas involving quantum computation, biological organisms, consciousness, non-locality and even what I called universal consciousness. These ideas are possibilities to be explored, not established scientific conclusions.

And perhaps that is exactly how they should be treated.

A hypothesis becomes scientifically interesting not because it sounds profound, but because we can ask:

What observation would prove it wrong?

The microtubule question

One of the most fascinating examples is the proposal that quantum processes involving neuronal microtubules could contribute to consciousness.

This idea has been associated particularly with the Orch-OR hypothesis and related approaches.

It is an attractive idea because microtubules are not merely passive structural elements. They are highly organized molecular structures inside cells, including neurons.

But attractiveness is not evidence.

The question is whether quantum coherence, entanglement or related phenomena can survive in the relevant biological environment long enough, and under the right conditions, to play a functional role in consciousness.

That is a much harder question.

Current research continues to discuss quantum and quantum-like approaches to neuroscience, but the empirical case for quantum consciousness remains unsettled. The 2026 literature itself emphasizes the importance of distinguishing experimentally demonstrated quantum effects from proposed explanations of consciousness.

For me, this makes the question more interesting—not less.

Because now we can ask what experiment would actually distinguish a quantum explanation from a classical one.

Perhaps the deeper question is not “quantum”

There is another possibility.

Maybe we have become too fascinated by the word *quantum*.

Perhaps the fundamental question is actually simpler:

What kind of organization is capable of experience?

Imagine two systems containing exactly the same molecules.

In one system, the molecules are arranged randomly.

In the other, they are organized into a dynamically interacting network with feedback, memory, energy flow and information exchange.

Chemically, the ingredients may be identical.

Functionally, the systems could be completely different.

So perhaps consciousness is not primarily about the ingredients.

Perhaps it is about the **relationships among the ingredients**.

This idea fits surprisingly well with the direction in which biology has been moving.

We increasingly think in terms of networks rather than isolated genes, pathways rather than isolated molecules, dynamics rather than static structures, and information flow rather than merely molecular inventories.

Could consciousness be another level of organization in this hierarchy?

I do not know.

But I think it is a question worth asking.

Could consciousness be a phase transition?

Here is where I would like to take the idea one step further.

We are familiar with phase transitions in physics.

Water behaves differently as a liquid and as ice. The difference is not because the molecules suddenly become different molecules. The difference arises from how they are organized.

Could something vaguely analogous happen in biological information processing?

Perhaps a system can increase its internal organization gradually. Information becomes more integrated. Feedback becomes stronger. Internal states begin influencing one another. The system develops increasingly sophisticated models of its surroundings—and eventually, perhaps, of itself.

At some point, could there be a qualitative transition?

Not simply **more information**,

but a different kind of information dynamics?

Not simply **more computation**,

but computation that includes the system's own state?

Not simply **response to the environment**,

but an internal representation of the relationship between the system and its environment?

And could that be where something resembling subjective experience begins?

This is speculation.

But it is a form of speculation that can potentially lead to questions we can investigate.

What would we actually look for?

If consciousness is a property of organization, then perhaps looking only for particular molecules will not be enough.

We might need to look for signatures of organization.

For example:

How strongly is information integrated?

How much of the system participates in a global state?

How quickly can information propagate?

How much memory does the system retain?

How does it distinguish internally generated states from external signals?

How does it maintain a model of itself?

How does its internal organization change with stress, development, disease or environmental change?

These questions connect naturally with the information-theoretic and systems-level perspective that we explored in the first articles of this series.

They also suggest something important.

Perhaps consciousness should not be treated only as a philosophical problem.

Perhaps at least some aspects of it can eventually become a **biological measurement problem**.

Could there be degrees of consciousness?

This possibility makes the question even more uncomfortable.

We tend to divide the world into two categories:

conscious and unconscious.

But biology rarely respects such neat boundaries.

There are gradients everywhere.

Temperature exists on a continuum.

Complexity exists on a continuum.

Information processing exists on a continuum.

Integration exists on a continuum.

Biological organization exists on a continuum.

So perhaps consciousness might also have degrees.

I am not suggesting that a bacterium necessarily has a tiny human-like consciousness.

That would be an unjustified leap.

Rather, I am asking whether the capacity for experience might emerge gradually as systems acquire particular forms of organization.

If that were true, consciousness would not necessarily be something that suddenly appeared at one particular evolutionary moment.

It might be more like a landscape.

Some systems would lie far outside it.

Others might occupy increasingly complex regions of it.

And humans might simply represent one particularly elaborate point on that landscape.

But there is a danger here

There is an obvious temptation in this kind of discussion.

Once we start seeing information processing everywhere, we may begin calling everything intelligent.

Once we see organization everywhere, we may start calling everything conscious.

That would make the idea meaningless.

A scientific concept must exclude as well as include.

If every organized system is conscious, then consciousness explains nothing.

Therefore, the challenge is not to expand the definition of consciousness until everything qualifies.

The challenge is to identify **specific properties of organization that could, in principle, distinguish conscious from non-conscious systems.**

That is much harder.

But it is also much more interesting.

Where does experience begin?

And so we return to the question in the title.

Where does experience begin?

At the neuron?

At the network?

At the brain?

At the organism?

Or does the question itself need to be reformulated?

Perhaps experience is not produced by one particular component.

Perhaps it is a property of a sufficiently integrated, self-referential, dynamically organized system.

Perhaps quantum effects contribute.

Perhaps they do not.

Perhaps consciousness requires something that neither classical information theory nor present-day quantum theory has yet captured.

We simply do not know.

And I think saying “**we do not know**” is scientifically healthier than pretending that we do.

From molecules to mind

When I look back at the journey of this series, I realize that we have travelled much further than I initially expected.

We started with molecules.

Then came information.

Then cellular organization.

Then computation.

Then biological intelligence.

Then collective intelligence.

And now we have arrived at consciousness.

At first glance, these may seem like completely different subjects.

Perhaps they are not.

Perhaps they are different layers of the same story:

matter → organization → information → computation → adaptation → intelligence → self-reference → experience.

I am not presenting this as an established biological sequence.

It is a conceptual map—a question about whether there is a deeper continuity connecting these levels.

If such a continuity exists, it could change how we think about life.

It would mean that the story of biology is not simply the story of increasingly complicated molecules.

It could also be the story of increasingly sophisticated **organization of information**.

And somewhere along that journey, something extraordinary may have happened.

Matter began not merely to react.

It began to **respond**.

Systems began not merely to respond.

They began to **adapt**.

Some systems began not merely to adapt.

They began to **coordinate**.

Some became intelligent.

And somewhere, somehow, experience appeared.

The next rethink

There is one final question that follows almost inevitably.

If intelligence can exist without a brain, and if consciousness might ultimately depend on a particular kind of information processing and organization, then another possibility opens up.

Could a non-biological system ever become conscious?

Could a sufficiently complex artificial system merely *simulate* consciousness?

Or could it actually *experience* something?

And if one day we build such a system, how would we know the difference?

That may be the next great boundary in the journey of *Rethinking Biology*.

Because perhaps the most difficult question is no longer:

“What is life?”

Perhaps it is:

“What does it mean for something to experience being alive?”

And that is where biology begins to meet neuroscience, artificial intelligence, physics—and philosophy.

The journey continues.

Scientific note

The established part of this discussion is that quantum mechanics is fundamental to the physical basis of biology and that quantum phenomena are being investigated in several biological processes. The stronger claim that quantum processes explain consciousness remains an open research question. Current work in quantum biology explicitly emphasizes the need for experiments that distinguish genuinely functional quantum effects from classical explanations.

Similarly, the ideas of a **consciousness continuum, universal consciousness, consciousness vectors, or quantum consciousness** should be understood as hypotheses and conceptual possibilities rather than established scientific theories. Their value, if any, will ultimately depend on whether they lead to precise, testable predictions.

That distinction is important to the philosophy of this entire series:

Rethinking biology does not mean replacing evidence with imagination. It means allowing imagination to ask questions that evidence can eventually answer.